



YOUCHENG LAN

Prospective PhD / RA Applicant
AI for Finance, Digital Finance, and On-chain Markets
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EDUCATION

Shandong University

Second-year Undergraduate Student

Current
Jinan, China

Major: Social Sport (Social Sports Guidance and Management); Minor: Finance

Academic preparation: AI for finance and blockchain; financial LLM agents; reinforcement learning and multi-agent systems; financial time series; market microstructure; volatility forecasting; DeFi and automated market making.

RESEARCH EXPERIENCE

Center for Artificial Intelligence and Digital Finance, School of Management, Shandong University

Chief Research Assistant

Current
Jinan, China

- Conduct hands-on research in financial AI, on-chain markets, LLM agents, MARL, trading benchmarks, and time-series modeling.
- Work on literature review, research framing, data collection, experiment automation, model evaluation, result interpretation, and manuscript writing.

WORKING PAPERS AND MANUSCRIPTS

DEXTwin: A Data-Calibrated LLM Agent Simulator for Mechanism Design in Decentralized Exchanges

AAAI 2027 targeted submission; first-author working paper (completed draft)

Abstract: Builds a data-calibrated economic digital twin for AMM-DEXs, using BDI-structured LLM agents to reproduce heterogeneous trader, arbitrageur, and LP behavior under real market frictions.

Project work: Integrated CEX prices, DEX pool/reserve data, gas data, and news context; implemented the Uniswap V2-style simulator, agent action schema, validation metrics, and mechanism counterfactual setup for fee, subsidy, gas, and MEV experiments; organized experiment outputs into paper-ready tables and robustness checks.

Language-Guided Multi-Agent Reinforcement Learning for Robust Arbitrage-Aware Market Making in Decentralized Exchanges:

AAAI 2027 targeted submission; first-author working paper (completed draft)

Abstract: Studies whether LLM-derived market beliefs can improve MARL policies under non-stationary arbitrage opportunities, gas shocks, liquidity fluctuations, and cross-pair regime shifts.

Project work: Built the calibrated DEX simulator interface, rule baselines, market-summary and belief-label schemas, balanced belief datasets, student-belief distillation workflow, MAPPO-style smoke suites, robustness tests, cross-market generalization checks, and market-quality/risk metrics; prepared the experiment protocol linking language beliefs to trading rewards.

RiskArena: A Regime-Stratified Benchmark for Position-Aware Multi-Agent Trading

AAAI 2027 targeted submission; first-author working paper (completed draft)

Abstract: Formulates financial-agent evaluation as an executable portfolio task rather than buy/sell/hold prediction, with regime-conditioned scorecards and auditable decision traces.

Project work: Constructed the S&P 100 public-data benchmark stack, regime labeling, transaction-cost/slippage/liquidity execution layer, deterministic baselines, SEC/GDELT text cache, LLM-agent adapters, RA-score diagnostics, cost ablations, pairwise tests, and reproducibility materials; designed the benchmark outputs to support agent comparison across market regimes.

TracePnL: Counterfactual Provenance Attribution for Portfolio Decisions

AAAI 2027 targeted submission; first-author working paper (completed draft)

Abstract: Develops causal attribution tools that connect realized PnL, risk changes, and decision errors to evidence, memory, debate, and agent-role traces for portfolio decisions.

Project work: Implemented the provenance graph logger, counterfactual replay engine, Trading Attribution Score, Causal Rationale Gap labels, surrogate replay model, no-leak LLM/news event panel, role/news ablations, stability analysis, and expert case-study pack; structured the traces so failed trades can be explained by data source, reasoning step, and execution choice.

WORKING PAPERS AND MANUSCRIPTS CONTINUED

LAMM: Language-Augmented Multi-Agent Market Making under Semantic and Microstructure Stress

AAAI 2027 targeted submission; first-author working paper (completed draft)

Abstract: Evaluates language-conditioned quote control under semantic shocks, toxic flow, liquidity drought, execution delay, volatility, and synchronized-spread pressure.

Project work: Built the multi-role quote-plan schema, public market/news preflight, twelve-scenario stress suite, matched-seed replay protocol, rule/Avellaneda–Stoikov/oracle/RL-tuned baselines, bootstrap and paired tests, long-horizon diagnostics, and ablations on skew, size, and role components; summarized stress-specific failure modes for the manuscript.

RIFT: Point-in-Time Retrieval-Augmented Volatility Forecasting for Market-Making Risk Control

ICDM 2026 targeted submission; first-author working paper (completed draft)

Abstract: Combines point-in-time analogue retrieval, regime-aware multi-view memory, uncertainty-aware forecasting, and leakage audits for volatility and tail-risk prediction.

Project work: Designed the analogue memory and retrieval protocol, built daily/minute market panels, implemented rolling-origin experiments, retrieval ablations, leakage stress tests, calibration checks, and downstream market-making risk-control evaluation using volatility/liquidity forecasts; wrote the evaluation plan around information-time consistency and risk-aware deployment.

Regime-Conditioned Sparse Mixture-of-Experts for Multi-Frequency Financial Time Series:

ICDM 2026 targeted submission; first-author working paper (completed draft)

Abstract: Proposes a lightweight sparse-MoE model that routes daily and intraday signals to regime-specialized experts for volatility calibration under non-stationary markets.

Project work: Built the daily/hourly data pipeline, structural data QA, regime feature encoder, sparse top-k router, expert bank, rolling-origin QLIKE evaluation, dense/linear/sparse comparisons, expert-activation diagnostics, and paper figures/tables; inspected expert routing behavior to explain which regimes activate each specialist.

Look-Ahead is Cheating: An Empirical Study of LLMs in Real-Time Finance

CIKM 2026, under review; first-author working paper

Abstract: Tests whether LLM trading decisions remain useful when future information is removed and decisions are generated only from information available at the trading date.

Project work: Built the ETF trading decision system across multiple information modes and models, automated daily account-state and decision logs, generated transaction-cost backtests, calculated Sharpe/excess-return/win-rate/bootstrap statistics, and audited decision validity; separated future-information settings from truly real-time settings to make the empirical comparison defensible.

Financial Agents for Portfolio Management: A Single-Model Agent Harness for Weekly SSE50 Allocation

Chinese Journal of Management Science manuscript; submitted, current status: With Editor; first-author manuscript

Abstract: Studies whether single-model financial agents can integrate macro, fundamental, factor, and synthesis roles for weekly SSE50 portfolio management under a fixed information cutoff.

Project work: Built the harness framework, DataBus over five iFind Excel datasets, macro/fundamental/factor/synthesis agents, constrained portfolio optimizer, weekly execution loop, XAI/provenance reports, DeepSeek/Qwen comparison, cost sensitivity, disagreement analysis, and paper-ready figures/tables; maintained weekly experiment records and converted results into manuscript figures.

RESEARCH INTERESTS

- AI for finance: financial agents, trading benchmarks, portfolio backtesting, and interpretable model evaluation.
- Blockchain and DeFi markets: on-chain data, decentralized exchanges, AMM simulation, and market mechanism design.
- Financial time series: volatility forecasting, market-making risk control, and regime-aware empirical finance.

TESTS, QUALIFICATIONS, AND BACKGROUND

IELTS: 6.5 GMAT: in preparation

Former Modern Pentathlon Athlete, Shenzhen National First-Class Swimmer, China ABRSM Piano Grade 8